

Tetrapyrrole Photochemical Spectral Database (TPSD)

— With a focus on 325 Synthetic Chlorophylls from Lindsey Lab
— Developed at Boston University as part of BF768, Spring 2026,
Professor Gary Benson

NC STATE

Team 15

Xiaohe Jin

Iris Lee

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Supervisor: Jonathan S. Lindsey

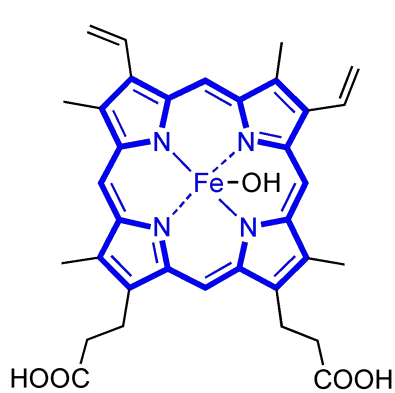
North Carolina State University

Department of Chemistry

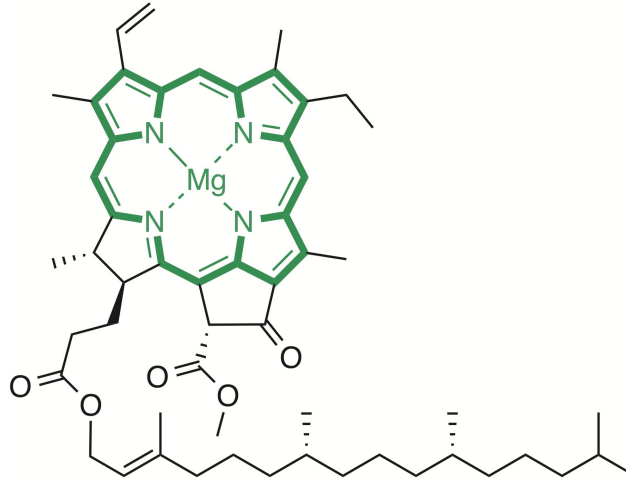
Expert collaborator: Dr. Masahiko Taniguchi

<https://chemistry.sciences.ncsu.edu/people/jslindse/>

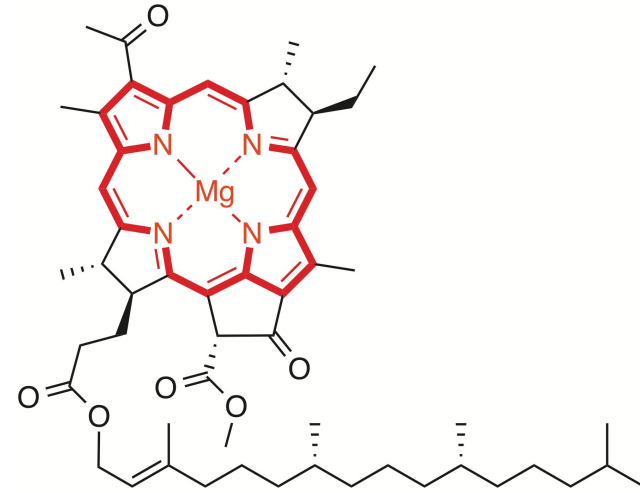
Introduction: Pigment of Life



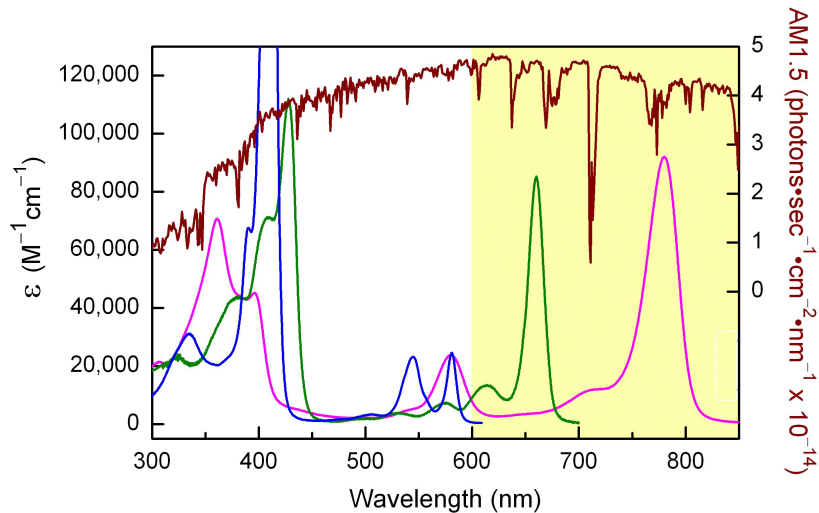
Hematin



Chlorophyll a



Bacteriochlorophyll a

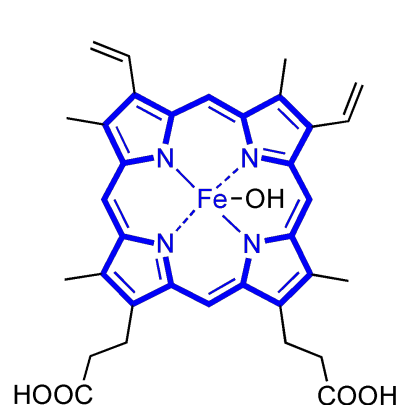


~1/2 of solar photons are >600 nm

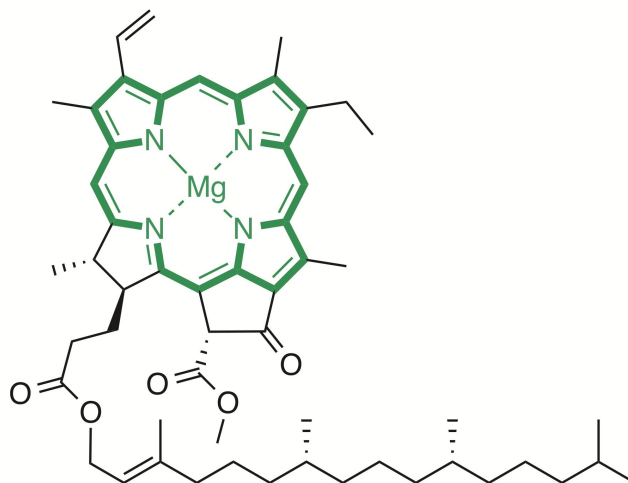


Tetrapyrrole macrocycles absorb strongly across the visible and near-infrared spectrum and thereby power the biosphere.

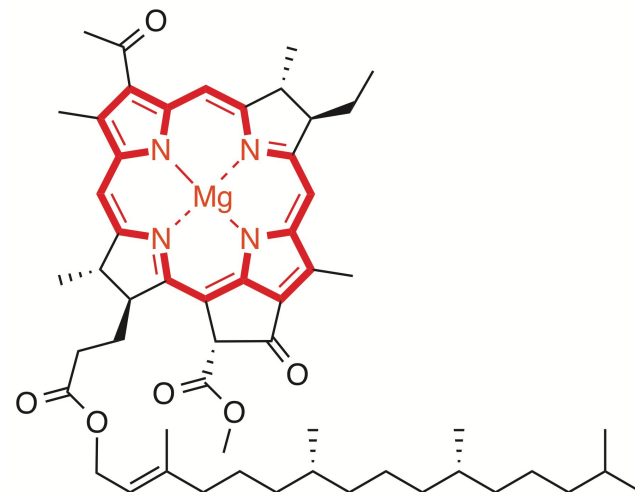
Introduction: *A synthetic platform technology*



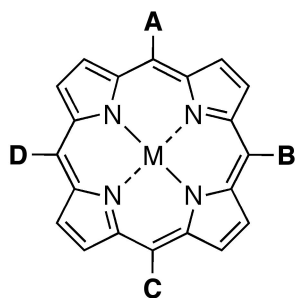
Hematin



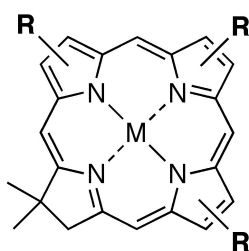
Chlorophyll a



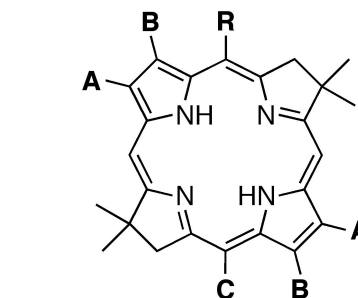
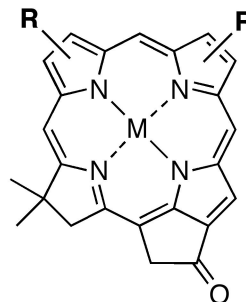
Bacteriochlorophyll a



ABCD-porphyrins
0-4 (non-identical)
substituents
1980–2005



Chlorins and 13¹-oxophorbins
*Can substitute each of the 12
peripheral sites independently*
1995–2015



Bacteriochlorins
*Can substitute each
site, still under
development*
2000–present

Synthetic building blocks of the native pigments.

Motivation and Project Aims

Motivation

- Synthetic chlorophyll derivatives are widely used to study how molecular structure affects absorption and fluorescence properties.
- However, the corresponding data are often distributed across raw lab files and literature tables rather than organized in a searchable system.
- This project addresses that gap by building a database for efficient storage, retrieval, and comparison of spectral data.

Aims

- Focus on synthetic chlorophyll derivatives from the Lindsey/Holten research effort.
- Build a structured database for synthetic chlorophyll compounds and their spectral properties.
- Implement wavelength-based searching to quickly identify compounds with desired photophysical features.
- Link metadata to raw spectrum files for downstream visualization and analysis.
- Create a user-friendly interface for exploring and comparing selected compounds.

Data Collection and Screening Workflow

2022 Lindsey lab
review article



Extract synthetic
chlorophyll entries and
spectral metadata



Match metadata with
lab-provided raw files



Select records with
usable spectrum files
and standardized fields



Load curated records
into database tables

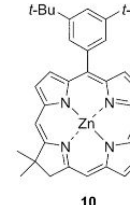
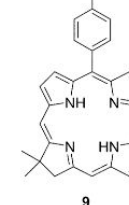
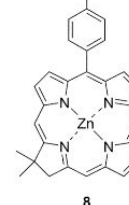
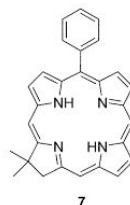
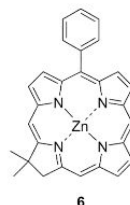
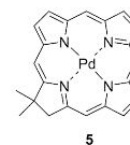
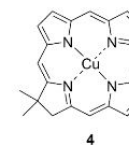
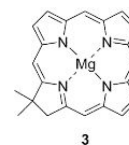
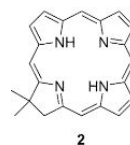
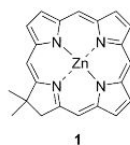
Beyond green with synthetic chlorophylls – Connecting structural features with spectral properties

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^c Department of Chemistry, Washington University, St. Louis, MO 63130-4889, USA



Structure data from the review

A	B	C	D	E	F	G	H	I	J
compound_id	file_name	file_path	solvent	absorption_k	molar_absor	instrument	measurement	reference_cc	investigator
1	L202 ZnC (SE197).abs.txt		toluene	398	240000	Agilent 8453	2006	L202	Lindsey lab
2	L212 H2C (SE211).abs.txt		toluene	389	158000	Agilent 8453	2007	L212	Lindsey lab
3	L212 MgC (SE212).abs.txt		toluene	402	224000	Agilent 8453	2007	L212	Lindsey lab
4	L211 CuC (SE204).abs.txt		toluene	397		Agilent 8453	2007	L211	Lindsey lab
5	L211 PdC.abs.txt		toluene	390		Agilent 8453	2007	L211	Lindsey lab
6	L311 ZnC-P5 (SE413).abs.t		toluene	404		Shimadzu 18	2013	L311	Holten lab
7	L311 FbC-P5 (SE409).abs.t		toluene	402		Shimadzu 18	2013	L311	Holten lab
8	L211 ZnC-T5 (SE203).abs.t		toluene	405		Shimadzu 18	2007	L211	Holten lab
9	L212 H2C-T5 (SE235).abs.t		toluene	403		Agilent 8453	2007	L212	Lindsey lab
10	L211 ZnC-B5 (SE220).abs.t		toluene	405	316000	Shimadzu 18	2007	L211	Holten lab

Metadata from Lindsey Lab

Website design - Home Page

Tetrapyrrole Photochemical Spectral Database (TPSD)

[Home](#) [Search](#) [Help](#)

Tetrapyrrole Photochemical Spectral Database (TPSD)

Database Overview

The Tetrapyrrole Photochemical Spectral Database (TPSD) is a specialized database designed to organize and provide access to chemical and spectral data for tetrapyrrole compounds synthesized in Dr. Jonathan S. Lindsey's laboratory.

Tetrapyrrole macrocycles include important biological pigments such as heme and chlorophylls and are widely used in photochemistry, materials science, and biological research. This database stores spectral and chemical information for synthetic chlorophyll derivatives.

Database Statistics

325

Compounds

324

Absorption Spectra

221

Emission Spectra

1

Compound Class

Database Features

- Search compounds by compound name
- Filter compounds by solvent
- Search by absorption wavelength range
- Search by emission wavelength
- Compare spectral properties across compounds
- Download spectral data and query results

Update Log

- **2026-04-28:** Uploaded 325 synthetic chlorophylls to TPSD.

- Overview of database, features, data sources

External Links

Literature

Beyond green with synthetic chlorophylls - Connecting structural features with spectral properties.
Taniguchi, M.; Bocian, D. F.; Holten, D.; Lindsey, J. S. J. Photochem. Photobiol. 2022, 52, 100513.

More Information

Visit the Lindsey Lab website for more information.
<https://sites.google.com/ncsu.edu/lindsey-lab/>

Data Source

The data included in this database originate from experimental measurements and literature related to tetrapyrrole compounds synthesized in the laboratory of Dr. Jonathan S. Lindsey.

Principal Investigator: **Jonathan S. Lindsey**
Department of Chemistry, North Carolina State University
Email: jslindse@ncsu.edu
Phone: 919.515.6406

Main Feature Overview / Search Page

- Search compounds by absorption wavelength range
- display matching compounds
- click one compound to open details
- show full spectrum plot

Search the Database

Use the form below to search for tetrapyrrole compounds by compound name, wavelength range, solvent, or fluorescence quantum yield.

Search Options

Compound Name:

Solvent:

Absorption Wavelength Min (nm):

Absorption Wavelength Max (nm):

Emission Wavelength Min (nm):

Emission Wavelength Max (nm):

Search

Clear

Search Results and Visualizations - Result table

10 result(s) found.

Compare	Compound ID	Compound Name	Solvent	Absorption λ_{max} (nm)	Emission λ_{max} (nm)	Quantum Yield	Details
☒	1	L202 Znc (SE197)	toluene	398.00	605.00	0.057	View Details
☐	2	L212 H2C (SE211)	toluene	389.00	634.00	0.2 (w/o O2)	View Details
☒	18	L212 H2C-M10 (SE225)	toluene	395.00	639.00	0.2 (w/o O2)	View Details
☐	20	L229 FbC-Pn10 (SE434)	toluene	394.00	637.00	0.18 (w O2)	View Details
☒	22	L212 H2C-P15 (SE251)	toluene	395.00	640.00	0.17 (w/o O2)	View Details
☐	24	L311 FbC-P20 (SE410)	toluene	393.00	639.00	0.21 (w/o O2)	View Details
☐	185	L212 H2C-Br15	toluene	396.00	640.00	0.0035 (w O2)	View Details
☐	214	L212 Oxo-H2C (SE298)	toluene	400.00	634.00	0.12 (w/o O2)	View Details
☐	303	L236 FbC1-PO3H2 (SE266)	PBS	398.00	628.80	-	View Details
☐	308	L236 FbC2-PO3H2 (SE267)	PBS	399.00	631.20	0.11 (w/o O2)	View Details

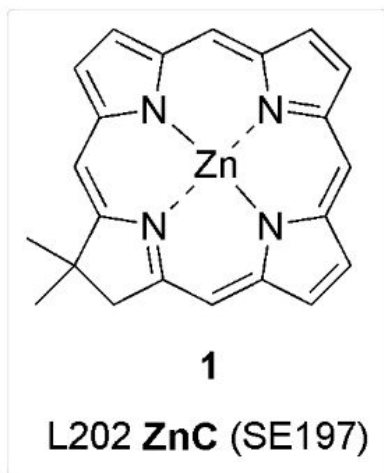
Compare (3)

Search Results and Visualizations - Compound Details

[← Back to Results](#)

Compound Details: L202 Znc (SE197)

Structure



Metadata

- Compound ID: 1
- Compound Name: L202 Znc (SE197)
- Lab Code: L202

Spectrum Information

- Solvent: toluene
- Emission λ_{max} : 605.00 nm
- Absorption λ_{max} : 398.00 nm
- Quantum Yield: 0.057
- Instrument: PTI QM-4/2003 SE
- Year: 2006
- Reference: L202
- Investigator: Lindsey lab

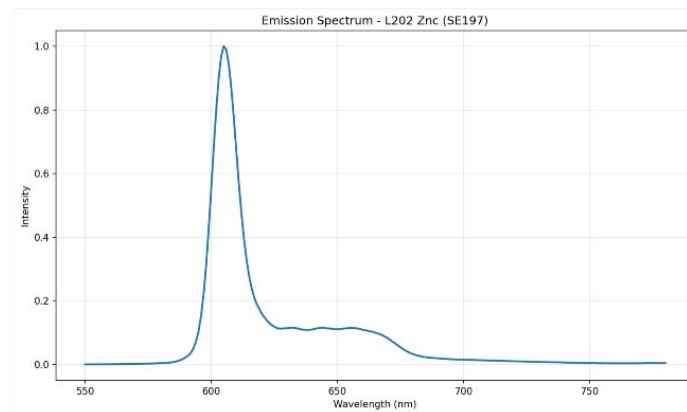
Files

[Download Fluorescence Data \(TXT\)](#)

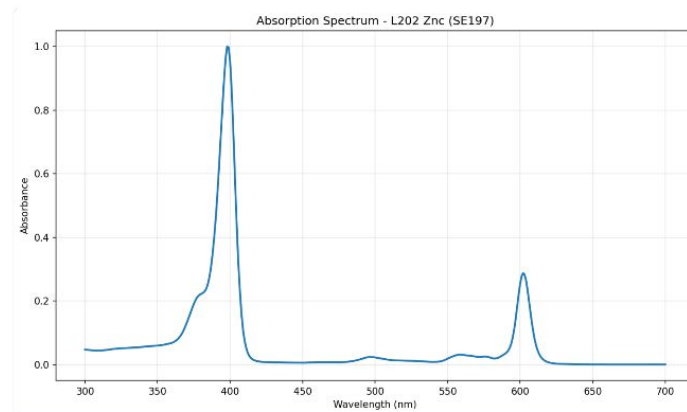
[Download Absorption Data \(TXT\)](#)

Spectrum Plots

Emission Spectrum



Absorption Spectrum



- structure image, metadata, spectrum information, links to raw download, view plots

Search Results and Visualizations - Compound Comparison

10 result(s) found.

Compare	Compound ID	Compound Name	Solvent	Absorption λ_{max} (nm)	Emission λ_{max} (nm)	Quantum Yield	Details
☒	1	L202 Znc (SE197)	toluene	398.00	605.00	0.057	View Details
☐	2	L212 H2C (SE211)	toluene	389.00	634.00	0.2 (w/o O2)	View Details
☒	18	L212 H2C-M10 (SE225)	toluene	395.00	639.00	0.2 (w/o O2)	View Details
☐	20	L229 FbC-Pn10 (SE434)	toluene	394.00	637.00	0.40 (w O2)	View Details
☒	22	L212 H2C-P15 (SE251)	toluene	394.00	637.00	0.40 (w/o O2)	View Details
☐	24	L311 FbC-P20 (SE410)	toluene	394.00	637.00	0.40 (w/o O2)	View Details
☐	185	L212 H2C-Br15	toluene	394.00	637.00	0.35 (w O2)	View Details
☐	214	L212 Oxo-H2C (SE298)	toluene	394.00	637.00	0.40 (w/o O2)	View Details
☐	303	L236 FbC1-PO3 (SE266)	toluene	394.00	637.00	0.40 (w/o O2)	View Details
☐	308	L236 FbC2-PO3 (SE267)	toluene	394.00	637.00	0.40 (w/o O2)	View Details

Compare (3)

Compare Compounds

1 - L202 Znc (SE197) Remove

18 - L212 H2C-M10 (SE225) Remove

22 - L212 H2C-P15 (SE251) Remove

Spectra to show

☒ Absorption

☒ Emission

Compare Spectra

Clear Selection

Compare (3)

Compare (3)

BOSTON UNIVERSITY

Search Results and Visualizations - Compound Comparison

[Back to Search](#)

Compare Compounds

Comparing 3 selected compound(s). Use the controls below to customize the Chart.js spectra view.

Selected Compounds

Compound ID	Compound Name	Lab Code	Absorption λ_{max} (nm)	Emission λ_{max} (nm)	Quantum Yield	Details
1	L202 Znc (SE197)	L202	398.00	605.00	0.057	View Details
18	L212 H2C-M10 (SE225)	L212	395.00	639.00	0.2 (w/o O2)	View Details
22	L212 H2C-P15 (SE251)	L212	395.00	640.00	0.17 (w/o O2)	View Details

Spectra View

Spectra to show

☒ Absorption
☒ Emission

Wavelength Min (nm):

Wavelength Max (nm):

Line Width:

2.0

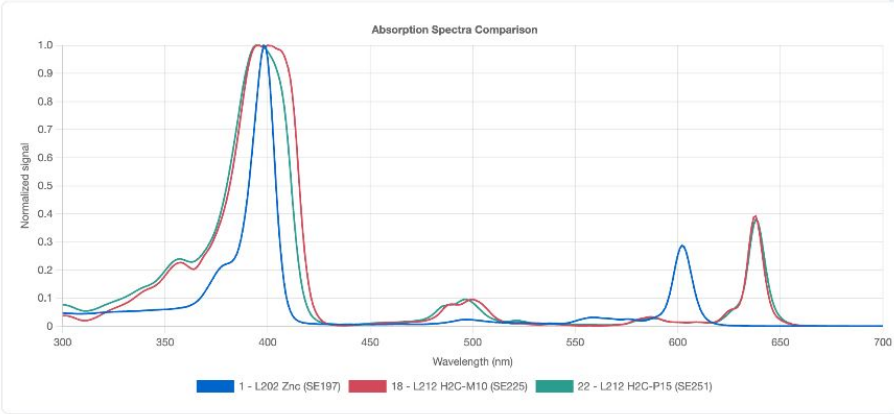
☒ Normalize curves

☒ Show grid

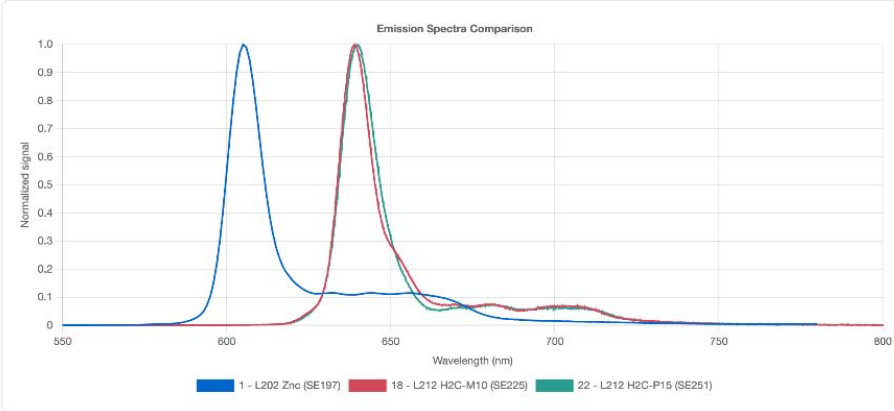
Update Plots

Reset Plot Settings

Absorption



Emission



Help Page

Tetrapyrrole Photochemical Spectral Database (TPSD)

[Home](#) [Search](#) [Help](#)

Help Page

Welcome to the Tetrapyrrole Photochemical Spectral Database (TPSD) help page. Click each section title below to expand or collapse the content.

1. About This Database ▲

The Tetrapyrrole Photochemical Spectral Database (TPSD) is designed to organize and provide access to chemical and spectral data for tetrapyrrole compounds. The database contains information related to compound identity, absorption spectra, fluorescence spectra, solvent conditions, fluorescence quantum yield, and literature references.

This database is intended to help users search, compare, and review spectral properties of compounds for research and educational purposes.

The home page also displays current database statistics, an update log, and external links to related literature and the Lindsey Lab.

2. What You Can Search

3. How to Search the Database

4. Example Searches

5. Understanding Search Results

6. Comparing Compounds

7. Compound Detail Pages

8. Downloading Data

9. Tips for Better Searching

10. Report an Issue ▲

If you encounter a problem while using the database, please use the form below to report the issue to the database team.

Name

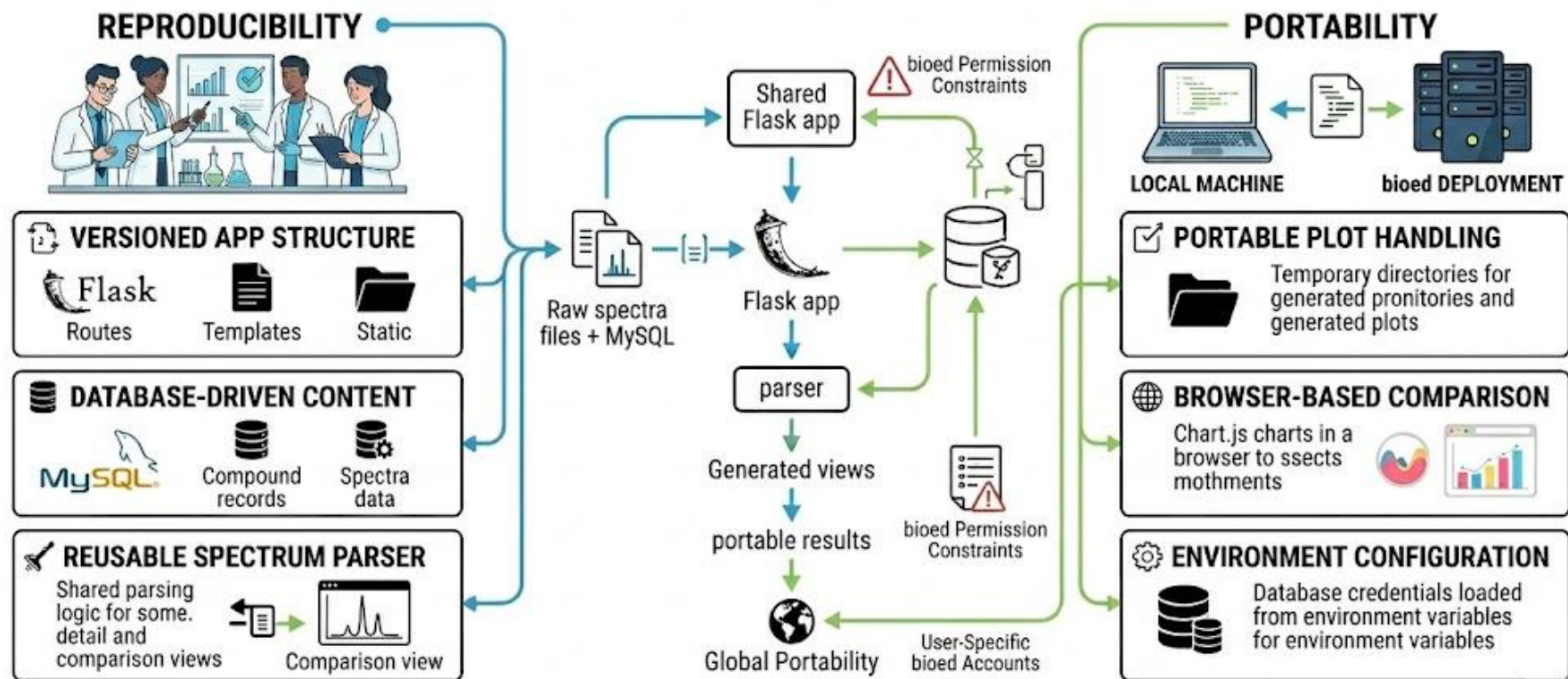
Email

Message

Submit

Reproducibility and Portability

DESIGNED FOR CONSISTENCY ACROSS TEAM MEMBERS AND DEPLOYMENT ENVIRONMENTS (LOCAL DEV & bioed)



Limitations

- Compounds belong to one class
 - Natural Chlorophylls, etc. could be added
- Only solvent as of now is toluene
- Exclusive to Lindsey lab
 - Needs more universal naming system

Future Work

- Expand database to include more compounds
- Classify compounds further
 - Based on substitutions, metal-chelated, etc.
- Machine learning and predictive modeling

Thank you!

Questions?